Impedance Matching – Part 2

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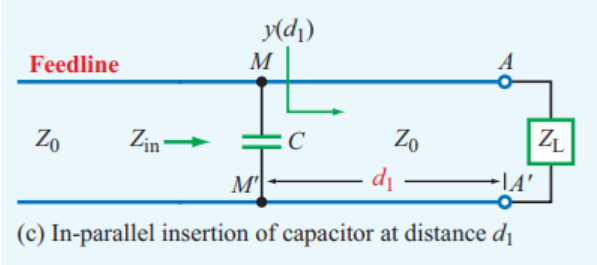
Lecture Objective

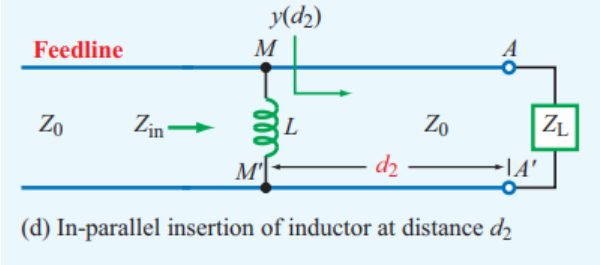
- By the end of this lecture, you will be able to:
  - Create lumped-element and single-stub MNs for matching a load impedance to a reference impedance.

Lumped-Element Matching

- Lumped-element matching makes use of a lumped element (capacitor or inductor) inserted in shunt at a distance from the load where the line admittance has a real part which is equal to the characteristic admittance. (Note that lumped elements could also be inserted in series, but such insertion is often impractical.)
- The value of the capacitor or inductor is chosen to cancel out the imaginary part of the line admittance.



(c) In-parallel insertion of capacitor at distance d_1



(d) In-parallel insertion of inductor at distance d_2

2. Lumped-Element Matching

- Solution:

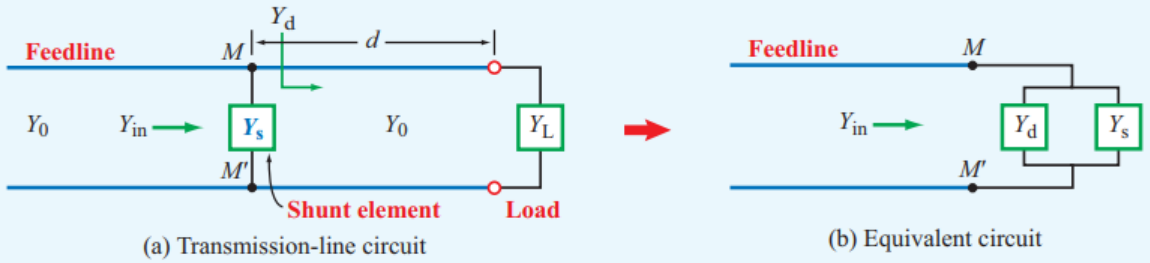


Figure 2-34 Inserting a reactive element with admittance Y_s at MM' modifies Y_d to Y_{in}.

$$Y_d = Y(d)$$

$$Y_{in} = Y_d + Y_s$$

For matching:

$$Y_{in} = Y_0$$

$$Y_s = jB_s$$

$$\Rightarrow Y_d = Y_0 - jB_s$$

Lumped Element MN: Design Equations

$$t = \begin{cases} \dfrac{X_L \pm \sqrt{R_L\left[\left(Z_0 - R_L\right)^2 + X_L^{\,2}\right]/Z_0}}{R_L - Z_0}, & R_L \neq Z_0 \\[2ex] \dfrac{-X_L}{2Z_0}, & R_L = Z_0 \end{cases}$$

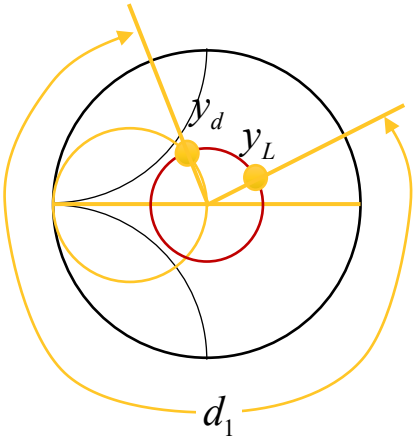
$$B_S = -\left\{\frac{R_L^{\,2}t - \left(Z_0 - X_L t\right)\left(X_L + Z_0 t\right)}{Z_0\left[R_L^{\,2} + \left(X_L + Z_0 t\right)^2\right]}\right\}$$

$$\frac{d}{\lambda} = \begin{cases} \dfrac{1}{2\pi}\operatorname{atan}\left(t\right), & t \geq 0 \\[2ex] \dfrac{1}{2\pi}\left(\pi + \operatorname{atan}\left(t\right)\right), & t < 0 \end{cases}$$

Lumped Element Matching

- Solution with shunt capacitor ($B_s > 0$):
- Move the normalized load admittance on a circle of constant reflection coefficient magnitude toward the generator until it intersects the g = 1 circle on the inductive (negative susceptance) side of the chart.

$$y_d = \frac{Y(d_1)}{Y_0} = 1 - jb_S$$



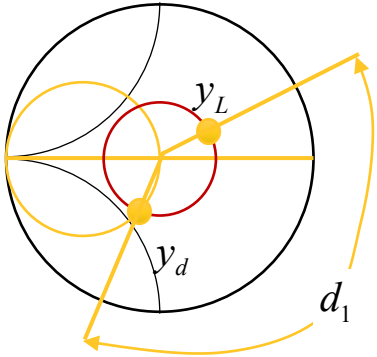
- Calculate the capacitance needed to cancel the imaginary part of the line impedance.

$$C = \frac{b_S Y_0}{\omega}$$

3. Lumped Element Matching

- Solution with shunt inductor ($B_s < 0$):
- Move the normalized load admittance on a circle of constant reflection coefficient magnitude toward the generator until it intersects the g = 1 circle on the capacitive (positive susceptance) side of the chart.

$$y_d = \frac{Y(d_1)}{Y_0} = 1 - jb_S$$



- Calculate the inductance needed to cancel the imaginary part of the line impedance.

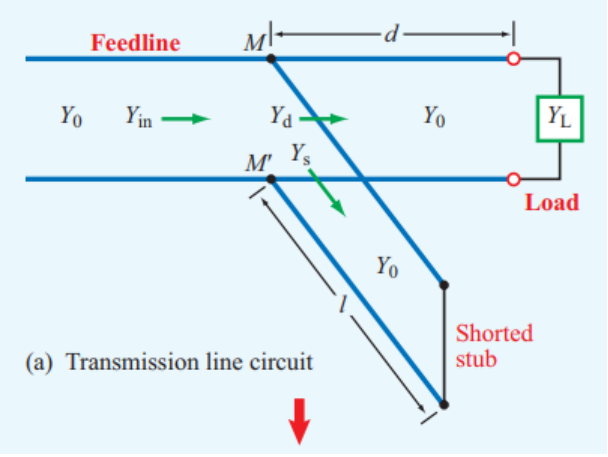
$$L = \frac{-1}{\omega b_S Y_0}$$

Single-Stub Matching

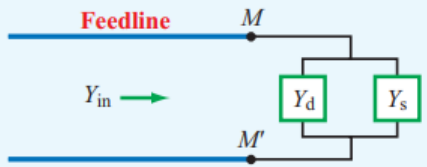
- Instead of inserting a lumped element in shunt to match the load to the feedline, a SC or OC stub can be used.
- Analytical equations can also be derived for obtaining these solutions. These derivations are given in the textbook.
- MATLAB routines which implement these equations have been developed and provided to the students of this class.

2. Single-Stub Matching

- Both SC and OC stubs can be used to realize MN.
- Shunt stubs are most common.



3. Single-Stub Matching



(b) Equivalent circuit

Figure 2-38 Shorted-stub matching network.

$$Y_d = Y(d)$$

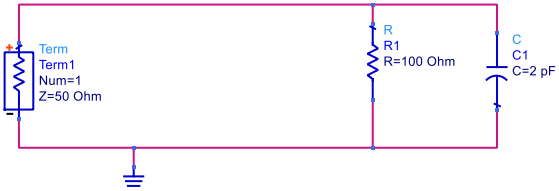
$$Y_{in} = Y_d + Y_s$$

For matching: $Y_{in} = Y_0$

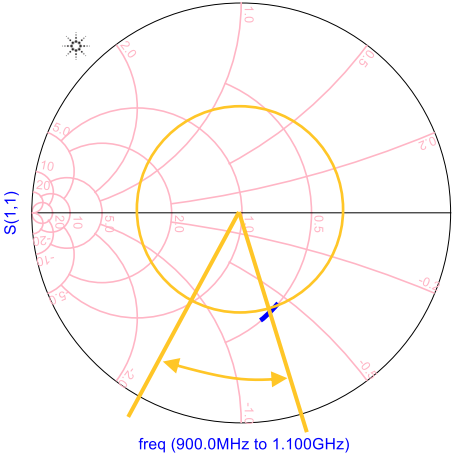
$$Y_s = jB_s = \begin{cases} -jZ_0 \tan(\beta l), & \text{SC stub} \\ jZ_0 \cot(\beta l), & \text{OC stub} \end{cases}$$

Single-Stub Matching Example

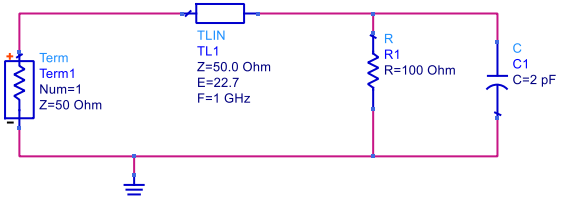
Obtain a perfect match at 1 GHz using a shunt stub, choose the solution with the shortest length of line between the load and the stub.



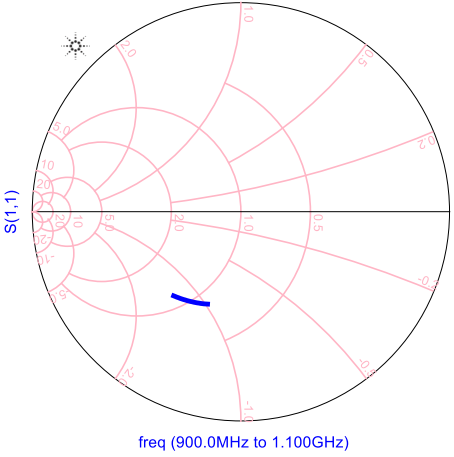
freq	S(1,1)
900.0 MHz	0.471 / -69.173
910.0 MHz	0.473 / -69.697
920.0 MHz	0.475 / -70.216
930.0 MHz	0.478 / -70.731
940.0 MHz	0.480 / -71.242
950.0 MHz	0.482 / -71.748
960.0 MHz	0.485 / -72.250
970.0 MHz	0.487 / -72.748
980.0 MHz	0.489 / -73.241
990.0 MHz	0.491 / -73.731
1.000 GHz	0.494 / -74.216
1.010 GHz	0.496 / -74.697
1.020 GHz	0.498 / -75.175
1.030 GHz	0.501 / -75.648
1.040 GHz	0.503 / -76.118
1.050 GHz	0.505 / -76.583
1.060 GHz	0.507 / -77.045
1.070 GHz	0.510 / -77.503
1.080 GHz	0.512 / -77.958
1.090 GHz	0.514 / -78.408
1.100 GHz	0.517 / -78.856



2. Single-Stub Matching Example



freq	Yin1
900.00 MHz	0.0173 + j0.0197
910.00 MHz	0.0175 + j0.0200
920.00 MHz	0.0178 + j0.0203
930.00 MHz	0.0180 + j0.0206
940.00 MHz	0.0183 + j0.0209
950.00 MHz	0.0186 + j0.0212
960.00 MHz	0.0188 + j0.0215
970.00 MHz	0.0191 + j0.0218
980.00 MHz	0.0194 + j0.0221
990.00 MHz	0.0197 + j0.0224
1.0000 GHz	0.0200 + j0.0227
1.0100 GHz	0.0203 + j0.0230
1.0200 GHz	0.0207 + j0.0234
1.0300 GHz	0.0210 + j0.0237
1.0400 GHz	0.0213 + j0.0240
1.0500 GHz	0.0217 + j0.0243
1.0600 GHz	0.0221 + j0.0247
1.0700 GHz	0.0224 + j0.0250
1.0800 GHz	0.0228 + j0.0253
1.0900 GHz	0.0232 + j0.0257
1.1000 GHz	0.0237 + j0.0260

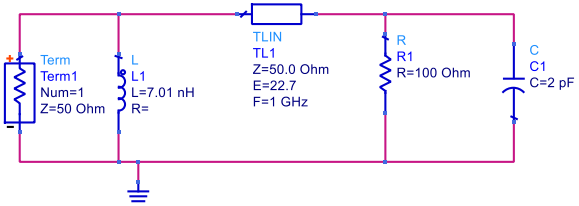


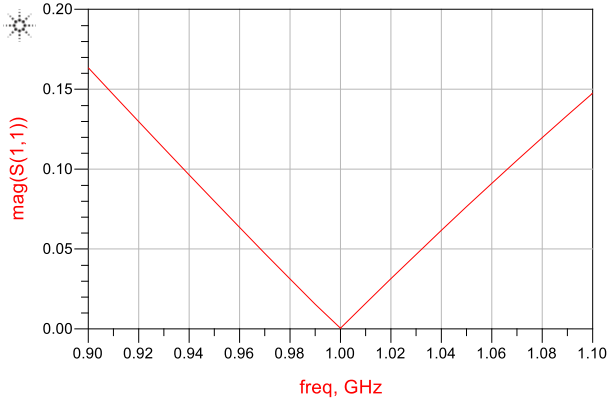
3. Single-Stub Matching Example

$$B_{stub} = -0.0227 \text{ S}$$ → inductor

$$B_{stub} = \frac{-1}{2\pi f L_{sh}} \rightarrow L_{sh} = \frac{-1}{2\pi f B_{stub}} = 7.01 \text{ nH}$$

4. Single-Stub Matching Example

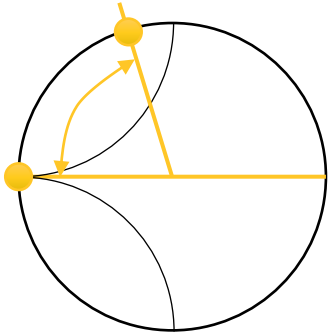




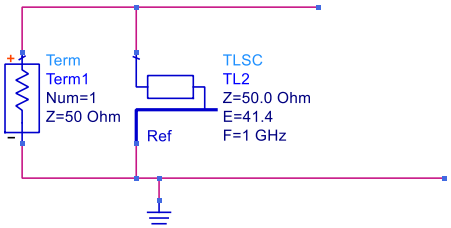
5. Single-Stub Matching Example

- Let's realize the shunt susceptance as a transmission line stub rather than a lumped inductor.
- Assume a short-circuited stub.

$$b_{stub} = -50 \times 0.0227 = -1.14$$

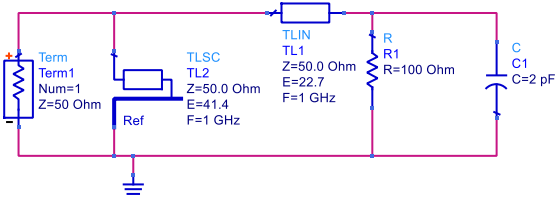


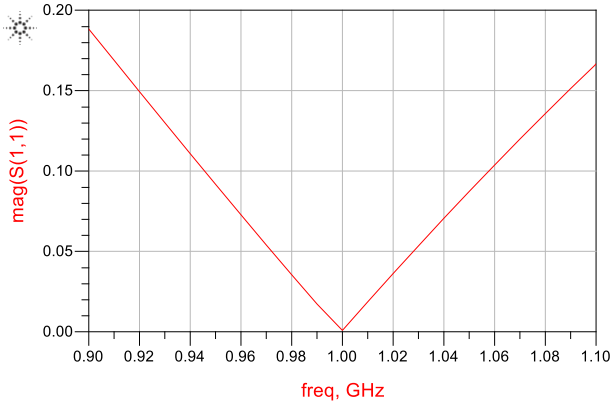
6. Single-Stub Matching Example



freq	imag(Yin1)
900.00 MHz	-0.0263
910.00 MHz	-0.0259
920.00 MHz	-0.0255
930.00 MHz	-0.0251
940.00 MHz	-0.0248
950.00 MHz	-0.0244
960.00 MHz	-0.0241
970.00 MHz	-0.0237
980.00 MHz	-0.0234
990.00 MHz	-0.0230
1.0000 GHz	-0.0227
1.0100 GHz	-0.0224
1.0200 GHz	-0.0220
1.0300 GHz	-0.0217
1.0400 GHz	-0.0214
1.0500 GHz	-0.0211
1.0600 GHz	-0.0208
1.0700 GHz	-0.0205
1.0800 GHz	-0.0202
1.0900 GHz	-0.0199
1.1000 GHz	-0.0196

7. Single-Stub Matching Example





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